

Entropy is not Enough for Test-Time Adaptation: From the Perspective of Disentangled Factors

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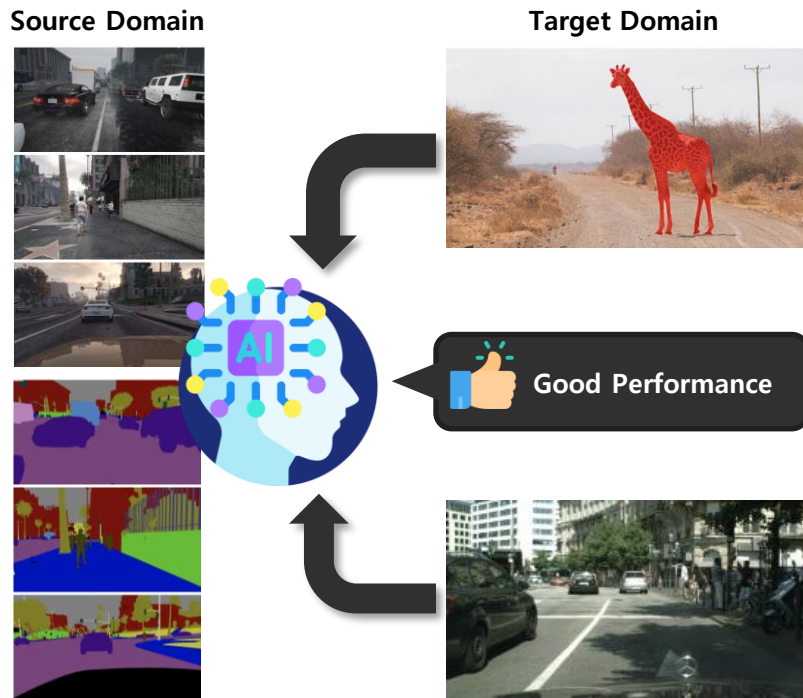
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01

Introduction

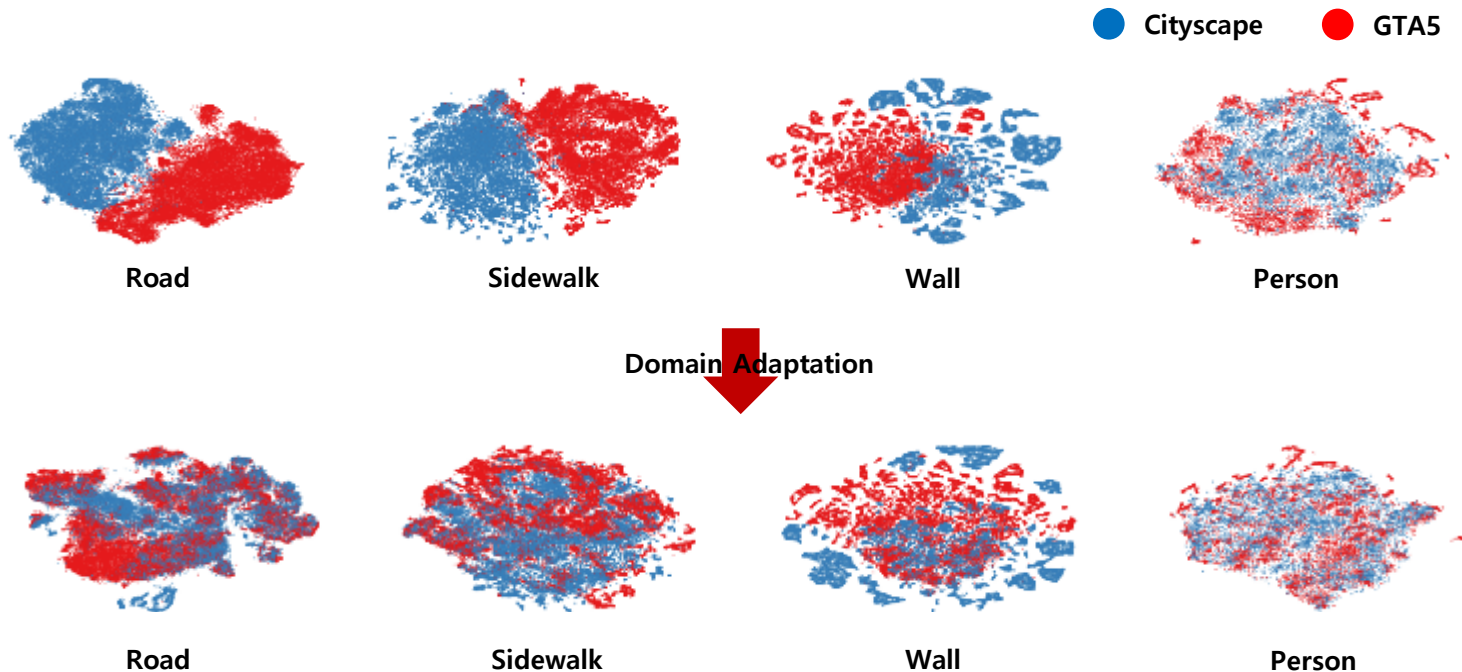
01. Introduction

1. Domain Adaptation



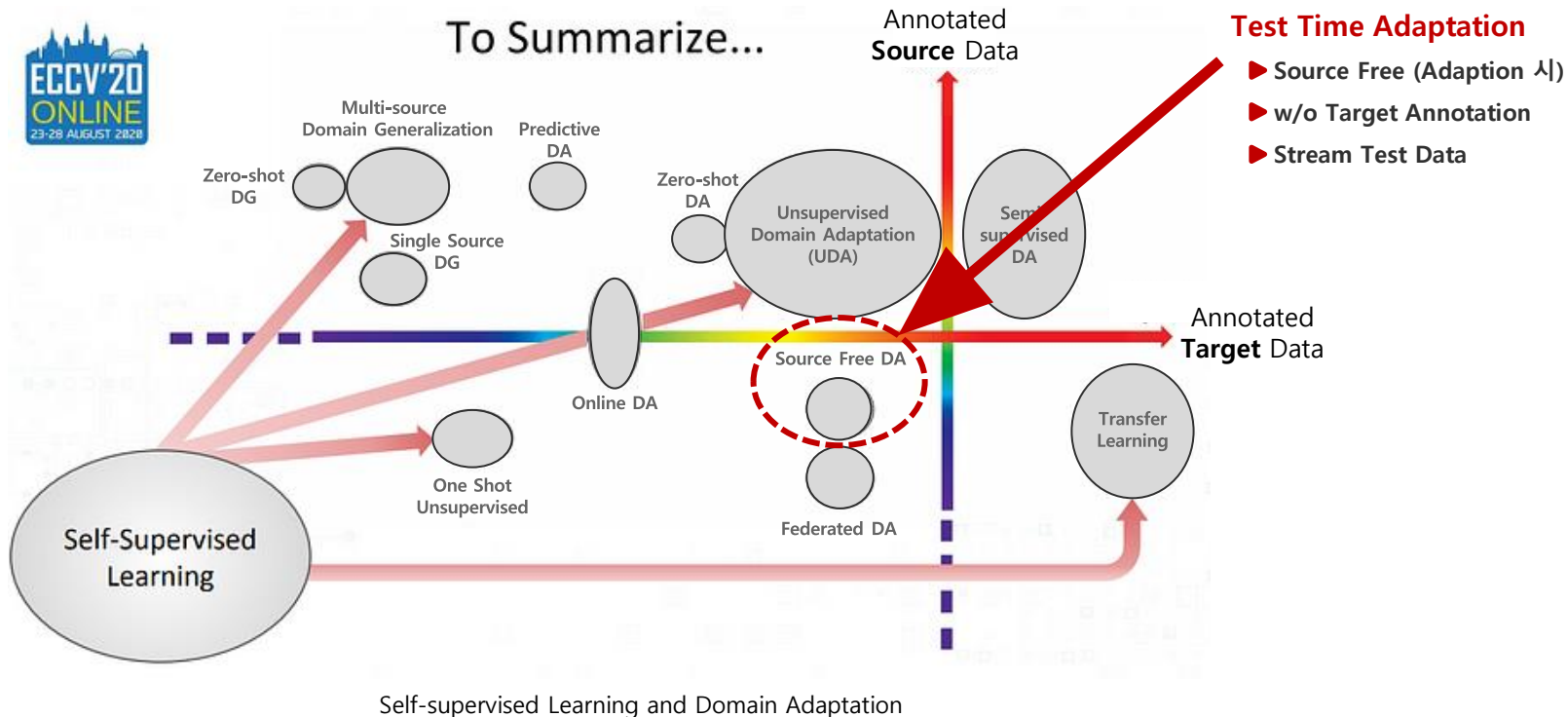
01. Introduction

1. Domain Adaptation



01. Introduction

2. Test-Time Adaptation



01. Introduction

2. Test-Time Adaptation

1 Stream Test Dataset

- 전체 Test Dataset의 분포를 알 수 없음
- 많은 양의 데이터가 지속적으로 입력됨
- ▶ 지속적으로 입력되는 데이터 중 어떤 데이터가 학습에 유효할 데이터인지 판단 필요
- ▶실 환경에서는, 중복 데이터 다량 발생

2 Source Free, w/o Target Annotation

- Source Data에 대한 접근 불가
- Target Annotation 없음
- ▶ Pseudo Label 또는 Surrogate Loss Function 등 필요

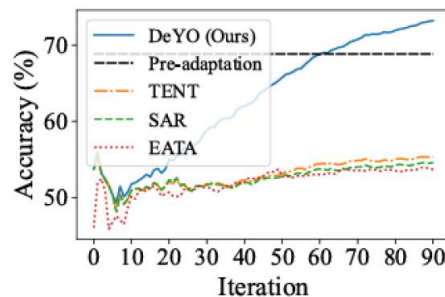


Figure 1: The accuracy within the worst group of the Waterbirds benchmark.

02

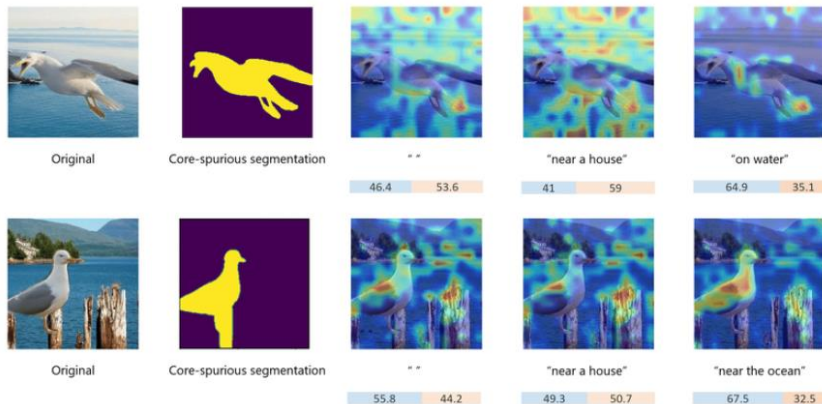
Previous Work

02. Previous Work

1. Motivating Observation

▪ Water Bird Dataset

- Domain Imbalance, Spurious Relationship을 연구하기 위한 데이터셋
- Class : Waterbird / Landbird 2종



Water Bird Dataset

- ▶ Group 1 : Waterbird on Water Background
- ▶ Group 2 : Waterbird on Land Background
- ▶ Group 3 : Landbird on Water Background
- ▶ Group 3 : Landbird on Land Background

02. Previous Work

1. Motivating Observation

▪ General Confidence Score : Entropy

- Waterbird Dataset 실험 결과, Entropy Level이 낮은 데이터들이 Accuracy가 가장 낮음

▶ Entropy는 신뢰도가 낮은 지표

$$\text{Ent}_{\theta}(\mathbf{x}) = -\mathbf{p}_{\theta}(\mathbf{x}) \cdot \log \mathbf{p}_{\theta}(\mathbf{x}) = -\sum_{i=1}^C \mathbf{p}_{\theta}(\mathbf{x})_i \log \mathbf{p}_{\theta}(\mathbf{x})_i$$

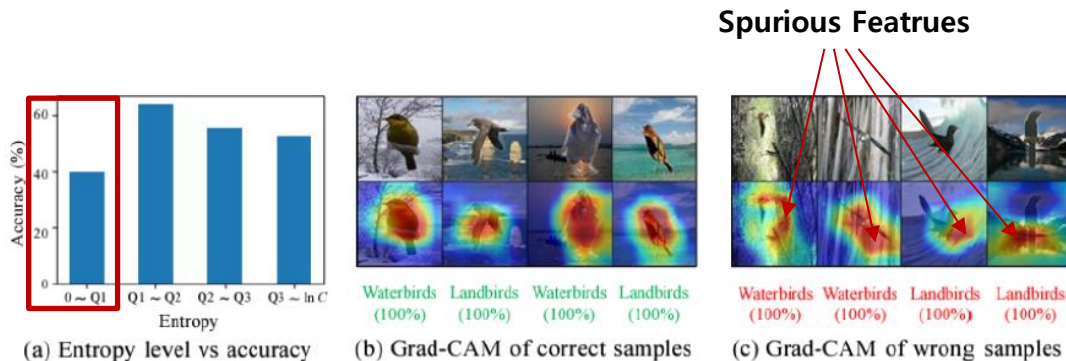


Figure 2: (a) A graph that represents accuracy by entropy levels. The lowest entropy interval 0~Q1 exhibits the lowest accuracy. (b) and (c) display Grad-CAM visualization of samples with correct and incorrect predictions with extremely low entropy, respectively.

02. Previous Work

2. Entropy is not Enough

- **Disentangled Latent Vector** : $\mathbf{V} = \{v_1, v_2 \dots v_{d_v}\}$
 - Input Image를 구별짓는 독립적인 Latent Vector
- **Correlation between Disentangled Vector and Label** : $corr_i^{train} = corr(y^{train}, v_i^{train})$
- 관련 연구¹⁾에 Motivation을 받아 Disentangled Vector를 4 종류로 나눔

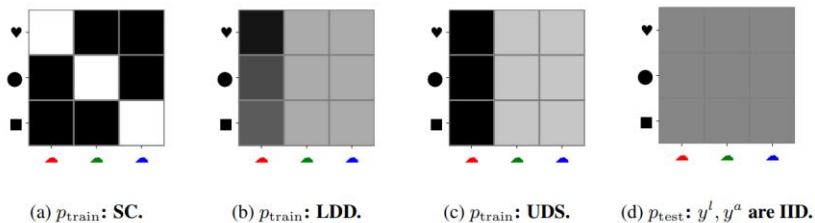


Figure 1: Visualization of the joint distribution for the different shifts we consider on the DSprites example. The lighter the color, the more likely the given sample. figure 1a-1c visualise different shifts over $p_{train}(y^l, y^a)$ discussed in 2.2: *spurious correlation* (SC), *low-data drift* (LDD), and *unseen data shift* (UDS). figure 1d visualises the test set, where the attributes are uniformly distributed.

$$\mathbf{v}_{pp} = \{v_i | corr_i^{train} > 0, corr_i^{test} > 0\}, \quad \mathbf{v}_{pn} = \{v_i | corr_i^{train} > 0, corr_i^{test} \leq 0\}$$

▶ **Train / Test 둘 다 Positive**
(Pos. : Label과 GT가 높은 관계성)

▶ **Train : Positive, Test : Negative**
(Pos. : Label과 GT가 높은 관계성)
(Neg. : Label과 GT가 낮은 관계성)

$$\mathbf{v}_{np} = \{v_i | corr_i^{train} \leq 0, corr_i^{test} > 0\}, \quad \mathbf{v}_{nn} = \{v_i | corr_i^{train} \leq 0, corr_i^{test} \leq 0\}$$

▶ **Train : Negative, test : Positive**
(Pos. : Label과 GT가 높은 관계성)
(Neg. : Label과 GT가 낮은 관계성)

▶ **Train : Negative, test : Negative**
(Neg. : Label과 GT가 낮은 관계성)

02. Previous Work

2. Entropy is not Enough

- **Low Entropy Sample Can be Harmful Sample**

- 4 Partition of Network Parameter : $\theta = \{\theta_{pp}, \theta_{pn}, \theta_{np}, \theta_{nn}\}$, corresponding to $\{v_{pp}, v_{pn}, v_{np}, v_{nn}\}$

- Logits of Linear classification : $a_{\theta}(x) = \theta \cdot V(x) = \theta_{pp} \cdot v_{pp} \dots$

- pseudo-label : $\hat{y} = \begin{cases} 1 & (a_{\theta}(x) > 0) \\ -1 & otherwise \end{cases}$

- Adaptation 초기에는, Entropy 기반의 high confidence를 가지는 $\hat{y} = 1$ 에 대해 아래 수식 만족

$$a_{\theta}(x) = \theta_{pp} \cdot v_{pp} + \theta_{pn} \cdot v_{pn} + \theta_{np} \cdot v_{np} + \theta_{nn} \cdot v_{nn} \gg 0, \\ |\theta_{pp} \cdot v_{pp} + \theta_{pn} \cdot v_{pn}| \gg |\theta_{np} \cdot v_{np} + \theta_{nn} \cdot v_{nn}|.$$

- 결론적으로 Train Dataset에서 학습 된 모델은 v_{pp}, v_{pn} 이 Dominant Factor가 되어있고, v_{np}, v_{nn} 은 0에 가까움

- v_{pp} : Commonly Positively-correlated with label

- ▶ **CPR Factor**

- v_{pn} : TRAIIn-time only Positively-correlated with label

- ▶ **TRAP Factor**

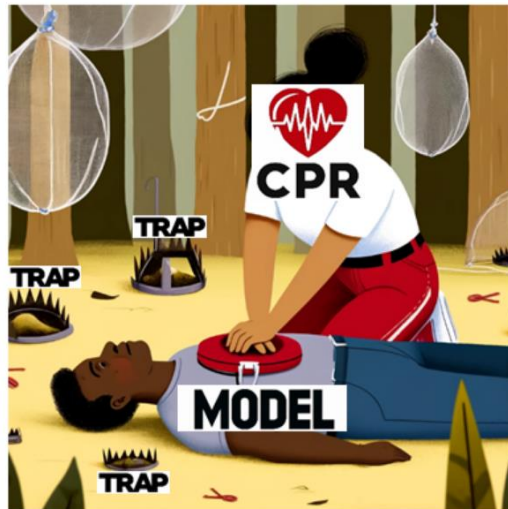
03

Methods

03. Methods

1. Strategy

- Entropy는 Disentangled Factor를 반영하지 못함
 - ▶ Entropy를 대체하는 Pseudo-Label Probability Difference (PLPD) Score 제안
- Foreground의 특징인 CPR Factor를 강화하고 TRAP Factor는 억제하는 TTA Method, Destroy Your Object (DeYO) 제시
 - ▶ 1) Sample Selection
 - ▶ 2) Parameter Weighting



All sources are created by DALL-E

03. Methods

2. Sample Selection

- 객체정보를 파괴하고 배경정보를 이용하여 TRAP Factor를 억제한 PLPD Score 계산 가능
 - 객체정보 파괴를 통한 CRP Factor 제외



Figure 3: Examples of a transformed image $\mathbf{x}'$ created by different object-destructive transformation methods. $\mathbf{x}$ is an example of WaterBirds.

- Original 이미지와 CRP Factor를 제외한 이미지를 빼 PLPD Score 계산

$$\begin{aligned}\text{PLPD}_{\theta}(\mathbf{x}, \mathbf{x}') &= (\mathbf{p}_{\theta}(\mathbf{x}) - \mathbf{p}_{\theta}(\mathbf{x}'))_{\hat{y}} \\ &= \underline{(\sigma(\boldsymbol{\theta} \cdot \mathbf{v}) - \sigma(\boldsymbol{\theta} \cdot \mathbf{v} - \boldsymbol{\theta}_{pp} \cdot \mathbf{v}_{pp}))}_{\hat{y}}\end{aligned}$$

- TRAP Factor를 최대한 억제한 Sampling

▶ Backpropagation 없이 Forward pass를 통해 계산되며 Overhead 최소화

$$S_{\theta}(\mathbf{x}) = \{\mathbf{x} | \text{Ent}_{\theta}(\mathbf{x}) < \tau_{\text{Ent}}, \text{PLPD}_{\theta}(\mathbf{x}, \mathbf{x}') > \tau_{\text{PLPD}}\}$$

03. Methods

3. Sample Weighting

- Selection 된 Sample을 통해 Update

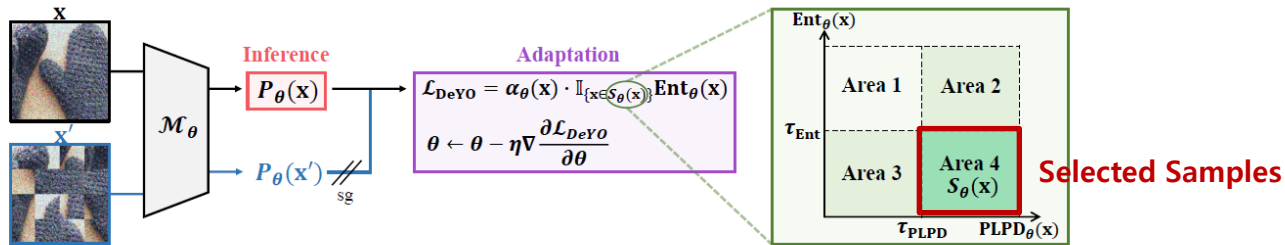


Figure 4: The overview of DeYO. DeYO comprises sample selection (Sec. 3.1) and sample weighting (Sec. 3.2) mechanisms. The areas within the green box are distinguished based on entropy and PLPD intervals, with Area 4 corresponding to $S_\theta(x)$ in Sec. 3.1.

- Update Weighting Function

▶ PLPD Term을 추가하여 Shape(CPR Factor)에 더 큰 가중치 부여

$$\alpha_\theta(x) = \underbrace{\frac{1}{\exp\{(\text{Ent}_\theta(x) - \text{Ent}_0)\}}}_{\text{Entropy Term}} + \underbrace{\frac{1}{\exp\{-\text{PLPD}_\theta(x, x')\}}}_{\text{PLPD Term}}$$

04

Experiments

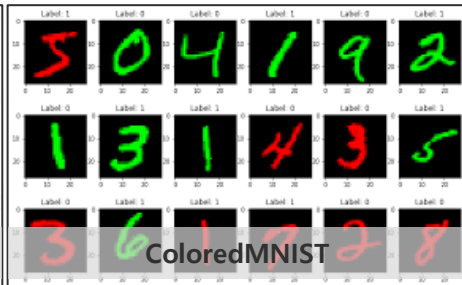
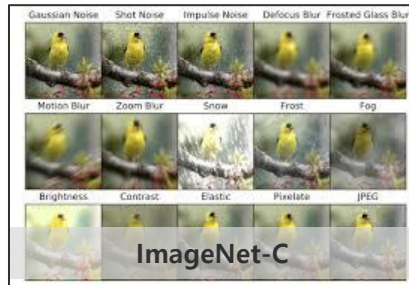
04. Experiments

0. Purpose

- 다양한 Benchmark와 실 환경과 유사한 Setting에서의 유효성 확인
- PLPD가 DeYO에서의 역할 확인
- 각 Components가 성능에 어떤 영향을 미치는가

1. Benchmarks

- **ImageNet-C**
 - 15 Corruption Types, 각 5단계 Severity Level
 - TTA 성능 평가를 위한 표준 benchmark
- **ColoredMNIST**
 - 숫자 색상과 Class 간 상관관계를 이용하여 Extreme spurious correlation 확인
- **Waterbirds**
 - Background와 Class 간 Spurious correlation 확인
- **ImageNet-R**
 - 다양한 Style Domain(Cartoon, sketch 등)으로 구성된 Distribution Shift 데이터셋
- **VisDA-2021**
 - Domain Style Shift와 같은 Challenging wile scnario를 포함한 데이터셋



04. Experiments

2. Test Scenario

- **Mild Scenario**
 - 단일 Corruption Type을 포함한 단순 환경
- **Wild Scenarios**
 - Dynamic Label Shifts : Test 데이터의 Label 분포가 Dynamic하게 변화
 - Batch Size 1 : Single Sample로 TTA 수행
 - 여러 Corruption Types가 혼합된 상황
- **Biased Scenarios**
 - Colored MNIST 및 WaterBirds에서 Spurious correlation shift가 극단적으로 존재

3. Models

- **ResNet Variant**
 - ResNet-18-BN, ResNet-50-BN (Batch Normalization 기반)
 - ResNet-50-GN (Group Normalization 기반)
- **VitBase-LN**
 - Vision Transformer 모델 기반
 - Layer Normalization 사용

04. Experiments

4. Main Results

▪ Mild Scenario

Table 1: Comparisons with baselines on ImageNet-C at severity level 5 under a mild scenario regarding accuracy (%). The **bold** value signifies the top-performing result.

Mild	Noise			Blur				Weather				Digital				Avg.
	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	
ResNet-50-BN	2.2	2.9	1.8	17.9	9.8	14.8	22.5	16.9	23.3	24.4	58.9	5.4	16.9	20.7	31.7	18.0
• MEMO	7.5	8.8	8.9	19.8	13.0	20.7	27.7	25.3	28.7	32.2	61.0	11.0	23.8	33.0	37.6	23.9
• Tent	29.2	31.2	30.1	28.1	27.7	41.4	49.4	47.2	41.5	57.7	67.4	29.2	54.8	58.5	52.4	43.1
• EATA	34.9	37.1	35.8	33.4	33.0	47.1	52.7	51.6	45.7	60.0	68.1	44.4	57.9	60.6	55.1	47.8
• SAR	30.6	30.6	31.3	28.5	28.5	41.9	49.4	47.1	42.2	57.5	67.3	37.8	54.6	58.4	52.1	43.9
• DeYO (ours)	35.6$\pm$0.2	37.9$\pm$0.1	37.1$\pm$0.1	33.8$\pm$0.2	34.1$\pm$0.2	48.5$\pm$0.1	52.8$\pm$0.1	52.7$\pm$0.0	46.4$\pm$0.1	60.6$\pm$0.0	68.0$\pm$0.1	46.1$\pm$0.1	58.4$\pm$0.1	61.5$\pm$0.1	55.7$\pm$0.1	48.6$\pm$0.0

▪ Biased Sceniaro

Table 2: Comparisons with baselines on ColoredMNIST regarding accuracy (%).

Biased	Avg Acc	Worst-Group Acc
ResNet-18-BN	63.40	20.05
• Tent	57.06	9.80
• MEMO	63.77	6.23
• SENTRY	63.23	15.78
• EATA	60.81	17.98
• SAR	58.37	12.36
• DeYO (ours)	78.24	67.39

Table 3: Comparisons with baselines on WaterBirds regarding accuracy (%).

Biased	Avg Acc	Worst-Group Acc
ResNet-50-BN	83.16	64.90
• Tent	82.95	54.14
• MEMO	82.34	50.47
• SENTRY	85.77	60.90
• EATA	82.38	52.38
• SAR	82.60	53.41
• DeYO (ours)	87.42	73.92

04. Experiments

4. Main Results

▪ Wild Scenario

- Mixed Shifts

Table 4: Comparisons with baselines on ImageNet-C at severity level 3 and 5 under a mixture of 15 corruption regarding accuracy (%).

Mixed Shifts	Level 5	Level 3
ResNet-50-GN	30.6	54.0
• MEMO	31.2	54.5
• Tent	34.2	33.1
• EATA	38.2	56.1
• SAR	38.3	57.4
• DeYO (ours)	38.6$\pm$1.3	59.2$\pm$0.05
VitBase-LN	29.9	53.8
• MEMO	39.1	62.1
• Tent	24.1	70.2
• EATA	56.4	69.6
• SAR	57.1	70.7
• DeYO (ours)	59.4$\pm$0.1	72.1$\pm$0.01

- Dynamic Label Shift / batch Size 1

Table 5: Comparisons with baselines on ImageNet-C at severity level 5 under online imbalanced label shifts (imbalance ratio = ∞) or under batch size 1 regarding accuracy (%).

Label Shifts	Noise			Blur				Weather				Digital				Avg.
	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	
ResNet-50-GN	17.9	19.9	17.9	19.7	11.3	21.3	24.9	40.4	47.4	33.6	69.3	36.3	18.7	28.4	52.2	30.6
• MEMO	18.4	20.6	18.4	17.1	12.7	21.8	26.9	40.7	46.9	34.8	69.6	36.4	19.2	32.2	53.4	31.3
• Tent	3.6	4.2	4.4	16.5	5.9	26.9	28.4	17.9	26.2	2.3	72.2	46.1	7.3	52.3	56.2	24.7
• EATA	25.7	28.6	24.8	18.5	19.6	24.1	28.4	35.3	33.0	41.2	65.2	33.3	28.0	42.4	43.1	32.7
• SAR	33.7	36.9	35.3	19.3	20.3	33.8	29.8	21.9	44.7	34.9	71.9	46.7	6.6	52.3	56.2	36.3
• DeYO (ours)	42.5$\pm$0.5	44.9$\pm$0.2	43.8$\pm$0.3	22.2$\pm$0.0	16.3$\pm$0.2	41.0$\pm$0.2	13.2$\pm$0.8	52.2$\pm$0.4	51.5$\pm$0.5	39.7$\pm$27.4	73.4$\pm$0.1	52.6$\pm$0.2	46.9$\pm$1.2	59.3$\pm$0.1	59.3$\pm$0.0	43.9$\pm$2.0
VitBase-LN	9.4	6.7	8.3	29.1	23.4	34.0	27.1	15.8	26.4	47.4	54.7	44.0	30.5	44.5	47.6	29.9
• MEMO	21.6	17.4	20.6	37.1	29.6	40.6	34.4	25.0	34.8	55.2	65.0	54.9	37.4	55.5	57.7	39.1
• Tent	33.9	1.8	27.2	54.8	52.9	58.6	54.3	12.4	11.7	69.7	76.3	66.3	59.6	69.7	66.6	47.7
• EATA	36.2	34.7	35.5	43.4	44.3	49.3	48.5	53.2	53.5	62.3	72.7	18.8	58.0	64.7	62.8	49.2
• SAR	42.3	34.9	44.1	50.0	50.5	55.6	53.1	59.7	47.2	66.2	75.2	50.3	60.1	67.3	65.0	54.8
• DeYO (ours)	53.5$\pm$0.5	36.0$\pm$25.2	54.6$\pm$0.8	57.6$\pm$0.2	58.7$\pm$0.2	63.7$\pm$0.1	46.2$\pm$18.7	67.6$\pm$0.1	66.0$\pm$0.1	73.2$\pm$0.2	77.9$\pm$0.1	66.7$\pm$0.1	69.0$\pm$0.1	73.5$\pm$0.1	70.3$\pm$0.2	62.3$\pm$1.7
Batch Size 1	Noise			Blur				Weather				Digital				Avg.
	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	
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• Tent	3.1	4.2	4.0	16.5	5.3	27.4	30.3	17.7	24.9	2.0	72.1	46.2	7.8	52.6	56.3	24.7
• EATA	24.8	27.9	25.8	17.9	17.3	28.7	29.3	44.7	44.4	40.2	71.0	44.5	27.0	46.8	55.6	36.4
• SAR	23.3	26.6	23.9	18.5	15.2	28.6	30.3	44.0	44.7	29.0	72.3	44.6	13.1	46.8	56.1	34.5
• DeYO (ours)	41.8$\pm$0.7	44.7$\pm$0.4	43.0$\pm$0.7	22.5$\pm$0.1	24.7$\pm$0.3	41.8$\pm$0.1	24.4$\pm$0.8	54.5$\pm$0.2	52.2$\pm$0.2	20.7$\pm$26.8	73.5$\pm$0.0	53.5$\pm$0.2	48.5$\pm$0.3	60.2$\pm$0.0	59.8$\pm$0.1	44.4$\pm$1.2
VitBase-LN	9.5	6.8	8.2	29.0	23.5	33.9	27.1	15.9	26.5	47.2	54.7	44.1	30.5	44.5	47.8	29.9
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• Tent	43.0	1.6	43.9	52.8	48.8	55.9	51.3	22.9	21.1	66.9	75.1	65.0	54.0	67.0	64.3	48.9
• EATA	32.2	26.7	30.3	43.8	40.1	47.7	42.6	35.7	43.4	60.8	65.6	61.1	46.5	60.5	58.2	46.3
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• DeYO (ours)	54.0$\pm$0.7	52.1$\pm$3.6	55.1$\pm$0.8	58.8$\pm$0.1	59.5$\pm$0.1	64.2$\pm$0.1	53.5$\pm$5.5	68.2$\pm$0.1	66.4$\pm$0.0	73.7$\pm$0.1	78.3$\pm$0.0	68.2$\pm$0.1	68.9$\pm$0.1	73.8$\pm$0.1	70.8$\pm$0.3	64.4$\pm$0.7

04. Experiments

4. Role and Effect of PLPD

- 기존 Entropy 기반 Confidence Metric은 Spurious Correlation Shift 상황에서 한계 존재
- PLPD는 CPR Factors에 기반한 신뢰도 평가 가능 및 이를 통한 Spurious Feature Dependency 제거 가능
- WaterBirds Benchmark : Risk-Coverage Curve**
 - Risk : Prediction Error Rate
 - Coverage : 선택된 Sample의 비율
- ImageNet-C Benchmark : Area 별 Accuracy 분석**
 - Area 1~3 : Entropy 또는 PLPD 중 하나만 충족
 - Area 4 : Low Entropy + High PLPD

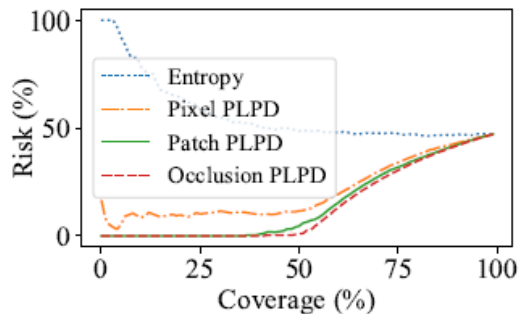


Figure 5: The Risk-Coverage curve of the worst group on Waterbirds.

Total 50,000 samples

Area 1 42,095 samples (84.2%) Acc : 9.9%	Area 2 1,513 samples (3.0%) Acc : 33.7%
Area 3 1,426 samples (2.9%) Acc : 33.5%	Area 4 4,966 samples (9.9%) Acc : 49.1%

Figure 6: Performance and percentage of identified samples on each area on ImageNet-C.

04. Experiments

5. Ablation

- Hyperparameter

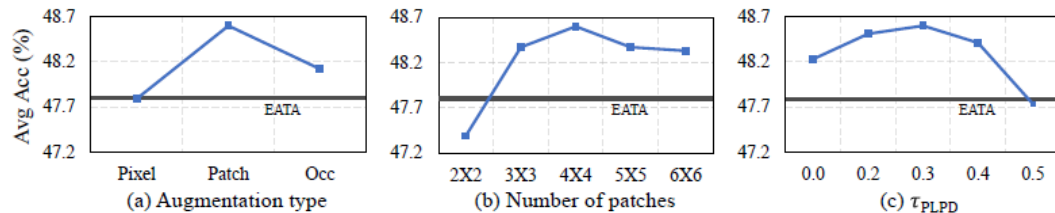


Figure 7: Hyperparameter experiments of DeYO on ImageNet-C at severity level 5.

- Components

Table 6: Ablation study on the proposed components. Second best = underline.

	$S_\theta(x)$		$\alpha_\theta(x)$		Avg.
	Ent _{θ}	PLPD _{θ}	Ent _{θ}	PLPD _{θ}	
(1) Tent					43.09
(2)				✓	44.92
(3)			✓		45.92
(4)			✓	✓	47.06
(5)		✓			44.78
(6) PLPD _{θ}		✓		✓	46.37
(7)		✓	✓		47.73
(8)		✓	✓	✓	<u>48.44</u>
(9)	✓				44.28
(10)	✓			✓	46.06
(11) Ent _{θ}	✓		✓		47.35
(12)	✓		✓	✓	47.61
(13)	✓	✓			44.68
(14)	✓	✓		✓	46.38
(15)	✓	✓	✓		47.95
(16) DeYO	✓	✓	✓	✓	<u>48.60</u>

05

Conclusion

05. Conclusion



Contribution

- ▶ 이론적 검증을 통한 Entropy 기반의 Confidence Metric 문제 증명
- ▶ Entropy의 문제를 해결하는 PLPD Score 및 TTA Method DeYO 제안
- ▶ 풍부한 실험을 통한 DeYO의 효과 증명



Limitation

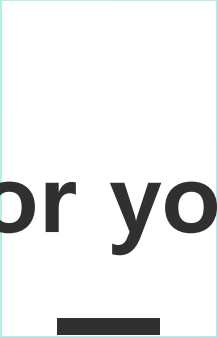
- ▶ 모든 종류의 CPR Factor를 확인할 수 없기 때문에 PLPD Score가 완벽하다고 할 수 없음
- ▶ Classification에 국한된 Method



Discussion

- ▶ Batch Size 1 실험에서 Test Data 입력 순서에 따른 성능 변화?
- ▶ Train Negative Correlation을 반영하지 않는 이유 ?
- ▶ 실 운용환경에 적용한다고 가정했을 때 Online 환경 구성 시, Inference-Train-Inference 3 Step 필요 ?
- ▶ Catastrophic Forgetting 문제는 발생하지 않을까?

Q & A



Thank you for your attention